

Week 23 & Week 24 Progress Summary

This week I finalised the model comparison and began building the final production-ready VerifAI detection pipeline. I consolidated evaluation results from all previously trained models, including audio-only, visual-only, temporal, and multimodal fusion systems, and compared them using accuracy, balanced accuracy, F1, ROC-AUC, average precision, and MCC.

Several experiments were conducted to improve reliability. The ViT visual baseline was retrained with lower learning rates, longer warmup, and threshold tuning, but it remained unstable and underperformed on CPU. The raw WAV encoder also showed high nominal accuracy but collapsed to majority-class prediction, which made it unsuitable as the final audio model. In contrast, the MFCC + ResNet18 audio model remained strong and stable.

On the visual side, I trained a new Xception-based model using frames from multiple datasets combined, including FaceForensics++ C23, Celeb-DF-v2, DeeperForensics, and FakeAVCeleb. This produced a much stronger and more generalisable visual backbone, achieving high overall accuracy, balanced accuracy, ROC-AUC, and MCC across datasets.

Using this improved visual backbone together with the MFCC audio model, I then trained a final hybrid audio-visual fusion system on FakeAVCeleb, which is the only dataset in the project containing both video and audio modalities. The final hybrid model achieved very strong performance, with accuracy above 99%, balanced accuracy above 99%, ROC-AUC close to 1.0, and a high MCC, making it the strongest and most balanced final system produced so far.

The final decision for deployment is to use the hybrid AV model when audio is available, and to fall back to the all-datasets visual Xception model when a video has no usable audio. This gives VerifAI a realistic real-world inference strategy for both videos with audio and videos without audio.